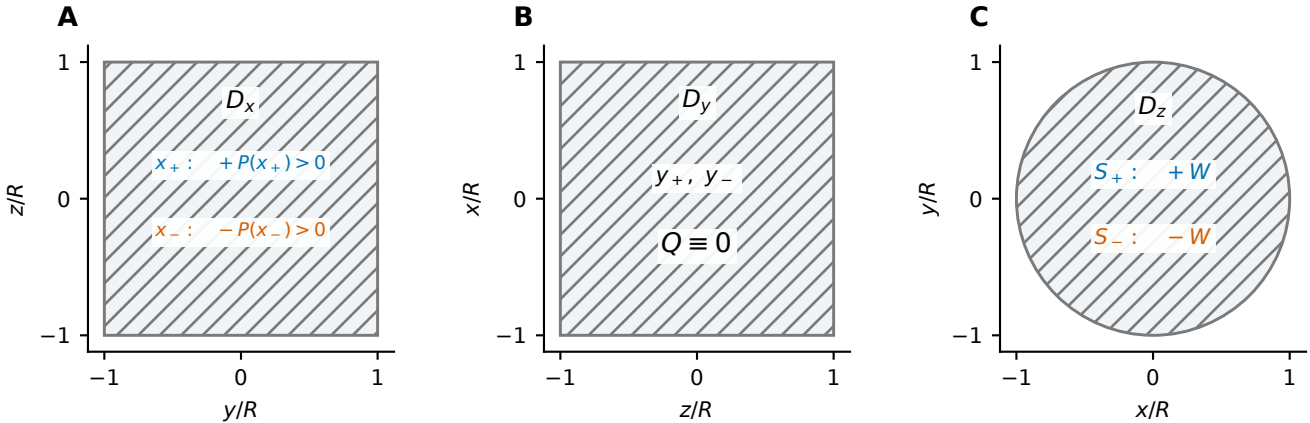

$I_x = \pi^2 R/2$

$I_y = 0$

$I_z = I_+ + I_- = 0$